ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 5

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 60

Marking guideline for the Academic Central Grade 5 Natural Sciences and Technology paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 5. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1 1.1 C ✓ 1.2 C ✓ 1.3 A ✓ 1.4 A ✓ 1.5 B ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 B ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 TRUE ✓ 2.2 TRUE ✓ 2.3 FALSE ✓ 2.4 FALSE ✓ 2.5 TRUE ✓ 2.6 TRUE ✓ (6)

[6]

QUESTION 3

3.1 Candle A ✓ (1)

3.2 Candle B ✓. The small jar holds the least air (oxygen), so it is used up first ✓ (2)

3.3 Air (oxygen) ✓ (1)

3.4 Switch off the stove and cover the pot with a lid or a damp cloth ✓ (never throw water on burning oil) (1)

3.5 Any TWO: the Sun (solar) ✓; wind ✓; moving water (hydro-electric); wood from new trees; waves (2)

3.6 Any TWO: coal is non-renewable and will run out ✓; burning coal pollutes the air ✓; it saves money; it helps to prevent load shedding (2)

3.7 Petrol or diesel ✓ (1)

[10]

QUESTION 4

4.1 Electricity (electrical energy) ✓ (1)

4.2 Any TWO: kettle ✓; iron ✓; stove; heater; toaster; geyser; hair dryer (2)

4.3 Advantage: cells can be carried anywhere / they are safer ✓. Disadvantage: they run out (go flat) / they cost a lot for the energy they give / they pollute when thrown away ✓ (2)

4.4 They contain chemicals that poison the soil and water ✓ (take them to a recycling point) (1)

4.5 Any TWO: people can be shocked or killed ✓; the wires can start fires ✓; children can touch the loose wires; it overloads the power lines and causes power cuts (2)

4.6.1 A rod (bar) through the centre of a wheel that the wheel turns on (or turns with) ✓ (1)

4.6.2 So that they roll smoothly (a round wheel turns easily) ✓ (1)

4.6.3 Any TWO: push or pull it ✓; wind up an elastic band ✓; a balloon; a spring; let it roll down a slope; a small motor with a cell (2)

4.6.4 So the wheels turn more easily (less rubbing / friction) ✓ (1)

4.7 Stored (potential) energy ✓ (1)

4.8 Input: energy from your hand (muscles) stored in the spring when you wind it ✓. Output: movement of the car (and some sound) ✓ (2)

[16]

QUESTION 5

5.1 Soil A ✓ (1)

5.2 Soil B ✓. Only 20 ml ran through, so it held back the most water ✓ (2)

5.3 100 ml − 20 ml = 80 ml ✓ (1)

5.4 Any TWO: the amount of soil ✓; the amount of water ✓; the size of the funnels; the time (10 minutes) (2)

5.5 Compost helps sandy soil hold water and gives it food (nutrients) for plants ✓ (1)

5.6 Any TWO: plant grass, trees or other plants ✓; do not let animals overgraze ✓; build small walls or terraces on slopes; cover the soil with mulch (2)

5.7 Igneous: granite, basalt or pumice ✓. Sedimentary: sandstone, shale, mudstone or limestone ✓ (2)

5.8 Any TWO: building houses and walls ✓; making roads ✓; making tools long ago; statues and gravestones; jewellery; making cement (2)

5.9 Any TWO: they tell us which plants and animals lived long ago ✓; they show how living things have changed ✓; they show how the land has changed (e.g. a sea where there is now land); they help to work out how old rocks are (2)

5.10 Any ONE: a burrow; a nest; animal droppings (dung); bite marks; tracks of a worm ✓ (1)

5.11 Any TWO: there is liquid water ✓; there is air (oxygen) to breathe ✓; it is not too hot or too cold; there is soil; light from the Sun (2)

[18]

TOTAL: 60

END OF MARKING GUIDELINE